

Optimizers in Deep Learning: Why Gradient Descent Needs Help

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1. Overview

Key point: An optimizer is the rule that turns gradients into weight updates. Plain gradient descent has five weak spots: choosing the learning rate, fixed learning-rate schedules, one learning rate for every direction, local minima and saddle points. The improved optimizers fix them with two ideas: build up speed, and adapt the learning rate.

Training a deep network can take a long time, so a lot of work goes into making it faster. Three techniques already help: good [weight initialisation](#), [batch normalisation](#) and the choice of [activation function](#). The fourth, and probably the most important for speed, is the **optimizer**: the algorithm that finds the values of the weights and biases for which the loss is smallest.

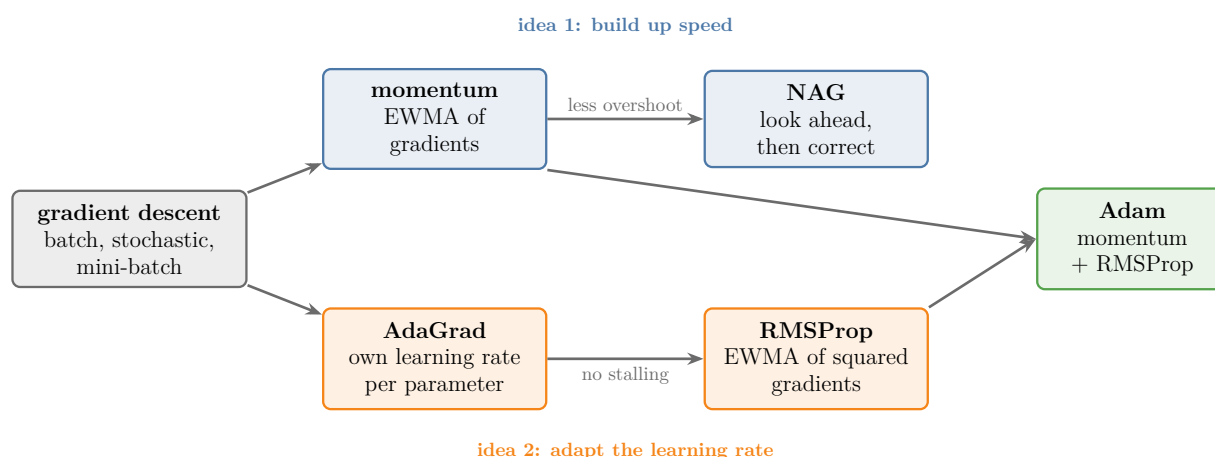


Figure 1: The optimizers ahead. Momentum and NAG build up speed; AdaGrad and RMSProp adapt the learning rate; Adam combines both

Figure 1 is the plan. Each optimizer is a small change to gradient descent, not something new from scratch. Several of them rely on one tool, the exponentially weighted moving average, which comes first.

2. Prerequisites

- The [backpropagation why Note](#): the loss as a function of all the weights, and the update rule.
- The [gradient descent in neural networks Note](#): batch, stochastic and mini-batch gradient descent in a network.
- The [gradient descent Note](#): the learning rate and the contour plot.

3. What an optimizer does

Key point: Training is an optimisation problem: find the weights and biases that make the loss smallest. We start from random values and improve them step by step.

Take a classification task with two input **features** (input variables, the columns of the data table) and one **target** (the output we predict). A network with one hidden layer of 2 nodes and one output node has $2 \times 2 + 2 = 6$ parameters in the hidden layer and $2 + 1 = 3$ in the output: 9 weights and biases in all (Notebook).

Training must find values of these 9 numbers for which the network's predictions $\hat{y}$ are as close as possible to the true values y . The loss measures the gap between y and $\hat{y}$, so training is an **optimisation problem**: minimise the loss. The weights and biases at the minimum are the network's best parameters.

The loss is a function of all 9 parameters, so its graph lives in 10 dimensions, which we cannot draw. With only 2 weights it becomes a surface over the (w_1, w_2) plane: we start at a random point and walk downhill to the lowest point, the **global minimum** (see section 3 of the [backpropagation why Note](#)).

4. Gradient descent, the optimizer so far

Key point: $w_{\text{new}} = w_{\text{old}} - \eta \nabla_w L$, repeated over many epochs. Batch, stochastic and mini-batch gradient descent differ only in how many rows they see before each update.

The optimizer used so far is gradient descent:

$$w_{t+1} = w_t - \eta \nabla_w L(w_t)$$

where η (eta) is the learning rate and $\nabla_w L$ the gradient of the loss with respect to the weights (see the [gradient descent Note](#)). We repeat the update for a chosen number of epochs.

Its three variants differ only in the number of rows used for each update (see the [gradient descent in neural networks Note](#)). With 500 rows and 10 epochs:

Variant	Rows per update	Updates in 10 epochs
Batch	all 500	10
Stochastic (SGD)	1	5,000
Mini-batch, batch size 100	100	50

So we already have three optimizers. The next section explains why we need more.

5. Five weak spots of plain gradient descent

Key point: Gradient descent is hard to tune (learning rate, schedule), treats every direction the same, and can get stuck at local minima and saddle points. Training then becomes slow, or ends at a poor solution.

These five challenges are the standard list for plain mini-batch gradient descent (Ruder 2016, §3).

5.1 Choosing the learning rate

Key point: Too small a learning rate converges painfully slowly; too large a one overshoots, oscillates or diverges. The right value depends on the data.

Every update subtracts η times the gradient, so η sets the size of every step (see section 5 of the [gradient descent Note](#)):

- **Too small:** every step is tiny and convergence is painfully slow; training may stop before reaching the minimum.
- **Too large:** the steps jump over the minimum, the path zigzags, and the loss can grow without limit.

A value in between works best, but finding it for a given dataset takes trial and error (Ruder 2016, §3).

5.2 Learning-rate schedules are fixed in advance

Key point: A schedule lowers the learning rate during training, but its timetable is fixed before training starts, so it cannot adapt to the dataset.

A **learning-rate schedule** changes the learning rate during training, usually lowering it after a set number of epochs or when the loss stops improving by a threshold (see section 6 of the [stochastic gradient descent Note](#)). The schedule and the thresholds must be defined before training. Different datasets need different schedules, so a schedule that works well on one dataset may fail on another (Ruder 2016, §3).

5.3 One learning rate for every direction

Key point: All parameters share one learning rate. When the loss is steep in one direction and flat in another, a rate safe for the steep direction is far too slow for the flat one.

With 9 parameters, the loss surface has 9 directions to move in. Gradient descent uses the same η for all of them, so it descends at the same “speed setting” in every direction. We cannot give w_1 one learning rate and w_2 another.

The loss is often very sensitive to some directions and insensitive to others (Goodfellow et al. 2016, §8.5). Take a narrow valley,

$$L(w_1, w_2) = \frac{1}{2} (w_1^2 + 100 w_2^2)$$

which is 100 times steeper across (w_2) than along (w_1). Each gradient descent step multiplies w_1 by $1 - \eta$ and w_2 by $1 - 100\eta$. So w_2 only shrinks if $|1 - 100\eta| < 1$, that is $\eta < 0.02$; with such a small η , w_1 shrinks by just 2% per step.

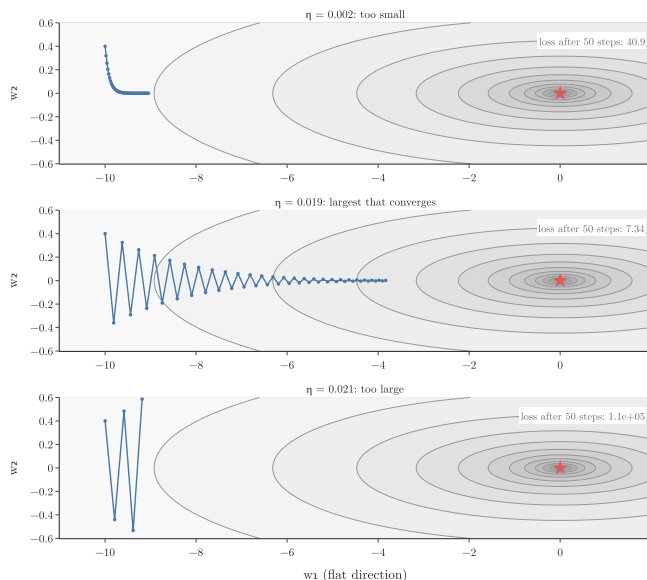


Figure 2: Gradient descent on the valley $L = (w_1^2 + 100w_2^2)/2$ from $(-10, 0.4)$, 50 steps each. Top: $\eta = 0.002$ barely moves along the valley. Middle: $\eta = 0.019$ zigzags across the valley and is still far from the minimum (star). Bottom: $\eta = 0.021$ bounces out of the valley

Figure 2 and the Notebook show the trap. Steps needed to bring the loss below 0.01:

η	0.002	0.005	0.01	0.019	0.02 or more
Steps	2,128	850	424	223	never

Even the best learning rate needs 223 steps, all because the steep direction caps η while the flat direction needs large steps. Plain gradient descent cannot go fast in one direction and slow in another.

5.4 Local minima

Key point: A network's loss has many dips. Gradient descent stops in the first one it reaches, which may be a local minimum with a worse loss than the global one.

The loss of a neural network is non-convex: it has several minima (see the [convex and non-convex cost functions Note](#)). The best solution is the **global minimum**, the weights

with the lowest loss of all. Gradient descent stops wherever the slope is zero, so starting from an unlucky point it settles in a **local minimum** and returns a sub-optimal solution. Stochastic gradient descent’s zigzag gives it some chance to jump out; batch and mini-batch gradient descent have less (see the [gradient descent in neural networks Note](#)).

5.5 Saddle points

Key point: At a saddle point the surface rises in one direction and falls in another. The slope there is zero and nearly zero on a wide plateau around it, so the updates almost stop.

A **saddle point** is a point where the surface slopes up in one direction and down in another (see section 3.3 of the [Hessian Note](#)). At the saddle itself the gradient is zero, so the update $w_{t+1} = w_t - \eta \times 0$ leaves the weights unchanged. Saddle points are usually surrounded by a plateau where the gradient is close to zero in every direction, so gradient descent crawls there for a long time even though the point is not a solution (Ruder 2016, §3).

Extra: For deep networks, saddle points may be a bigger obstacle than local minima: Dauphin et al. (2014) argue that in high dimensions most points with zero gradient are saddles, not minima (cited in Ruder 2016, §3; see also Goodfellow et al. 2016, §8.2.3).

5.6 What goes wrong as a result

Key point: Training is either very slow, or it ends at a poor solution.

Because of these weak spots, plain gradient descent either trains slowly (a tiny learning rate, a flat direction, a plateau) or stops at a worse solution than it could reach (a local minimum). The optimizers that follow are designed to do better on both counts.

6. The optimizers ahead

Key point: Momentum, NAG, AdaGrad, RMSProp and Adam. Each is a small change to the gradient descent update, and most of them use an exponentially weighted moving average.

Five optimizers improve gradient descent along the two ideas of Figure 1:

Optimizer	Main idea	Weak spot it targets
Momentum	keep a running average of past gradients and build up speed	flat directions, narrow valleys, small local dips
Nesterov accelerated gradient (NAG)	momentum that looks ahead before it corrects	momentum’s overshooting
AdaGrad	a separate, shrinking learning rate for every parameter	one learning rate for every direction

Optimizer	Main idea	Weak spot it targets
RMSProp	AdaGrad that forgets old gradients	AdaGrad's learning rate shrinking to nothing
Adam	momentum and RMSProp together	both

Momentum, RMSProp and Adam all keep an [exponentially weighted moving average](#) (EWMA): a running average that gives recent values more weight. We study the EWMA first.

Extra: Methods that use second derivatives, such as Newton's method, can also handle steep and flat directions (see the [Hessian Note](#)), but they are too expensive for the millions of parameters of a deep network (Ruder 2016, §4). The optimizers above use only the gradient.

7. Summary

Weak spot of gradient descent	What happens	Fixed by
Choosing the learning rate	too small: slow; too large: unstable	adaptive learning rates (AdaGrad, RMSProp, Adam)
Fixed schedules	cannot adapt to the dataset	adaptive learning rates
One learning rate for all directions	slow along flat directions of a narrow valley	momentum; per-parameter rates
Local minima	stops at a sub-optimal solution	momentum (can roll through small dips)
Saddle points	updates almost stop on the plateau	momentum, adaptive learning rates

- An optimizer finds the weights and biases that minimise the loss; gradient descent is the basic one.
- Batch, stochastic and mini-batch gradient descent differ only in the rows per update.
- The steep direction of a narrow valley caps the learning rate; the flat direction then crawls: 223 steps at best on our valley.
- The improved optimizers use two ideas: build up speed (momentum, NAG) and adapt the learning rate (AdaGrad, RMSProp); Adam uses both.
- Most of them rely on the exponentially weighted moving average.

8. Sources

- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*. MIT Press. §8.2.3 (saddle points), §8.5 (algorithms with adaptive learning rates).
- Ruder, S. (2016). An overview of gradient descent optimization algorithms. arXiv:1609.04747. §3 (challenges), §4 (algorithms).
- Dauphin, Y., Pascanu, R., Gulcehre, C., Cho, K., Ganguli, S. and Bengio, Y. (2014). Identifying and attacking the saddle point problem in high-dimensional non-convex optimization. NeurIPS 2014.

9. Key terms

Term	Meaning
Optimizer	The algorithm that turns the gradients into weight updates to minimise the loss, such as gradient descent or Adam
Optimisation problem	Finding the inputs (here the weights and biases) that make a function (here the loss) smallest

Term	Meaning
Feature	An input variable: one column of the data table
Target	The output we predict
Learning-rate schedule	A plan, fixed before training, for lowering the learning rate during training
Global minimum	The point with the lowest loss of all
Local minimum	A point lower than everything around it, but not the lowest overall
Saddle point	A point where the surface rises in one direction and falls in another; the gradient there is zero